

Rippl Project Requirements Document

Overview:

Rippl is a mobile app that lets users model any major life event as a financial scenario and see, in plain English, exactly what happens to their take-home pay, savings rate, tax burden, and long-term wealth. Users input their current financial situation paired with the financial information of a major life event such as “having a child”, “buying a home” or “a new job offer” etc and Rippl predicts what their financial situation will look like after that event has occurred before they decide to go through with it.

Problem Statement:

Users find it hard to determine what major decisions mean for their finances because spreadsheets are complex, calculators are too simple and financial advisors are too expensive which results in undesired future financial outcomes and underplanning.

Goals & Success Metrics:

- Users can complete the projection action in under 5 minutes
- Validation: 200 waitlist signups before build begins
- Month 1: 50 paying Pro subscribers within 30 days of public launch
- Month 9: 1,000 paying subscribers, 4.2+ App Store rating, <5% monthly churn
- Year 2: 10,000 paying subscribers, explore B2B/IFA white-label version

Retention Rate:

Day 1 retention: $\geq 60\%$

Day 7 retention: $\geq 35\%$

Day 30 retention: $\geq 20\%$

Month 3 retention: $\geq 15\%$

Daily Active Users/Monthly Active Users:

Target DAU/MAU ratio: ≥ 0.20 at Month 3, ≥ 0.30 by Month 9

Absolute DAU target by Month 3: ≥ 150

Absolute DAU target by Month 9: ≥ 800

Conversion Rate Optimization:

Waitlist $\rightarrow$ Download: $\geq 40\%$

Download $\rightarrow$ Account Created: $\geq 70\%$

Account Created $\rightarrow$ First Scenario Run: $\geq 60\%$

Free $\rightarrow$ Pro conversion: $\geq 8\%$ within 30 days of signup

Churned user $\rightarrow$ Reactivation: $\geq 10\%$ within 60 days

Error rate:

Application errors:

- Crash-free session rate: $\geq 99.5\%$

- API error rate: < 1% of all requests
- Scenario calculation failure rate: < 0.5%
- Open Banking connection failure rate: < 2%

AI output errors:

- AI summary generation failure rate: < 1%
- AI guardrail trigger rate: monitor and review any flagged outputs weekly

Time to First Value:

- Target TTFV: < 5 minutes

Non-Goals:

- Rippl will not give financial advice

Target Users/Personas:

- Decision makers (25-45) Professionals facing a major life event in the next 12 months — a job change, property purchase, or career pivot. They are comfortable with technology, underserved by generic calculators, and would pay for clarity. They do not want a financial adviser for every decision.
- Planners and financially anxious people. People who run financial scenarios for fun, or who feel anxious about money and want a tool that makes uncertainty feel manageable. This group drives word-of-mouth and app store ratings.
- Freelancers and self employed people. A growing segment with uniquely complex finances — irregular income, self-assessment tax, no employer pension. Mainstream apps ignore them. Rippl's freelance scenario module speaks directly to their needs.
- Young people who are financially conscious and are trying to figure out how much to save a month based on their own expenses. Rippl can help them plan their future and choose the jobs with the income they need for the life they want.

User Stories: (assumed - modify this using stakeholder interviews, user research, competitive analysis and analytics & feedback)

Authentication & Account Management

- As a new user, I want to create an account with a username and password, so that my data is personal and secure
- As a user, I want to reset my password, so that I can regain access to my account if I forget it
- As a user, I want to change my username, so that I can update how I'm identified in the app
- As a user, I want to delete my account, so that I can remove my data from Rippl entirely
- As a user, I want to log in with biometric authentication, so that I can access my account quickly and securely

- As a user, I want to be automatically logged out after inactivity, so that my financial data is protected if I leave the app open
- As a user, I want to manage my active sessions and log out of all devices, so that I can secure my account if my phone is lost or stolen
- As a user, I want Rippl to enforce a strong password policy, so that my account is protected from weak credentials

Profile Management

- As a user, I want to save my current financial information to my profile, so that Rippl can use it to generate accurate scenarios
- As a user, I want to upload a profile picture, so that my account feels personal
- As a user, I want to update all information in my profile except my email, so that I can keep my details current
- As a user, I want to connect my bank account via Open Banking, so that my financial profile is populated automatically and accurately
- As a user, I want to manually enter my financial information, so that I can use Rippl without connecting a bank account
- As a user, I want to see when my Open Banking data was last synced, so that I know how current my financial information is
- As a user, I want to manually trigger an Open Banking data refresh, so that I can ensure my profile reflects my latest transactions
- As a user, I want to disconnect my bank account, so that I can revoke Rippl's access to my banking data at any time

Onboarding

- As a new user, I want to be guided through setting up my financial profile before using scenarios, so that my first scenario is accurate and useful
- As a new user, I want to skip onboarding and complete my profile later, so that I can explore the app before committing my financial details
- As a new user, I want to choose between Open Banking and manual entry during onboarding, so that I can decide how I share my financial information

Scenario Engine

- As a user, I want to choose from a set of default scenarios, so that I can quickly model common financial decisions without building one from scratch
- As a user, I want to create a custom scenario, so that I can model financial situations specific to my circumstances
- As a user, I want to run up to three scenarios simultaneously, so that I can compare multiple financial outcomes at once
- As a user, I want to see a comparison table of my current financial figures against projected figures, so that I can clearly understand the financial impact of a scenario

- As a user, I want to receive an AI-generated summary alongside my comparison table, so that I can understand the implications of a scenario in plain English
- As a user, I want to see a confidence indicator on AI projections, so that I know how reliable the output is
- As a user, I want to see a disclaimer that AI outputs are not financial advice, so that I understand the limitations of Rippl's projections

Saved Scenarios

- As a user, I want to save a scenario with a custom name, so that I can identify and refer back to it easily
- As a user, I want to delete a saved scenario, so that I can keep my saved scenarios relevant and uncluttered
- As a user, I want to be notified when a saved scenario may be outdated due to profile changes, so that I know when to re-run it
- As a user, I want to duplicate a saved scenario, so that I can use it as a starting point for a variation without starting from scratch
- As a user, I want to add notes to a saved scenario, so that I can record my thoughts or context around it
- As a user, I want to share a saved scenario to an external app, so that I can discuss it with a partner, family member, or advisor

Projections & Visualisations

- As a user, I want to see a 12 month cash flow projection as a bar chart in a saved scenario, so that I can visualise how my finances will move month by month
- As a user, I want to see a 5 year projection view in a saved scenario, so that I can understand the long-term financial implications of a decision
- As a user, I want charts to include text-based alternatives, so that I can access the information if I use a screen reader

Reverse Mode

- As a user, I want to enter a target financial outcome and have Rippl tell me what finances I need to achieve it, so that I can plan backwards from my goals
- As a user, I want to see what changes to my current finances are required to meet a target outcome, so that I have a clear and actionable path forward
- As a user, I want to save a reverse mode result in the same way as a standard scenario, so that I can refer back to my goal planning

Notifications & Life Alerts

- As a user, I want to permit Rippl to send me notifications, so that I can stay informed about relevant financial events
- As a user, I want to turn on Life Alert notifications, so that I am notified when domestic or global events may impact my finances

- As a user, I want to choose which categories of Life Alert I receive, so that I only get notifications relevant to my situation
- As a user, I want to set a threshold for Life Alerts, so that I am only notified when the financial impact is significant enough to matter to me
- As a user, I want Life Alert notifications to link to relevant saved scenarios, so that I can immediately see how an event affects my specific financial plans

Family Mode

- As a user, I want to enable family mode, so that I can model scenarios based on joint finances with my partner or family
- As a primary user, I want to invite family members via email or link, so that they can contribute their financial data to shared scenarios
- As a primary user, I want to remove a member from my family group, so that I can manage who has access to our shared financial model
- As a family member, I want to leave a family group, so that I can disconnect my financial data from a shared model at any time
- As a user in family mode, I want to see income and expenses attributed to individual members, so that I can understand each person's contribution to the scenario

Accessibility & Personalisation

- As a user, I want to switch to dark mode, so that I can use Rippl comfortably in low light environments
- As a user, I want to switch back to light mode, so that I can revert to the default appearance
- As a user, I want to adjust text size, so that I can read Rippl comfortably regardless of my vision needs
- As a user, I want to navigate between tabs, so that I can move between different areas of the app quickly

Data & Privacy

- As a user, I want to export my data in a portable format, so that I can keep a personal record or transfer it elsewhere
- As a user, I want to request full deletion of my data, so that I can exercise my right to be forgotten under GDPR
- As a user, I want to accept Rippl's privacy policy and terms of service during signup, so that I understand how my data is used before I commit to it

Functional Requirements:

Authentication & Security:

- Users can create an account with a username and password (mvp)
- Users can reset their password (mvp)

- Users can change their username (mvp)
- Users will be able to log in with biometric authentication (Face ID / fingerprint) (post-mvp)
- Users will be able to manage active sessions and log out of all devices (post-mvp)
- Rippl will enforce a minimum password strength policy (mvp)
- Rippl will support two-factor authentication (2FA) (post-mvp)

Onboarding: (post-mvp)

- New users will be guided through an onboarding flow to populate their financial profile before using scenarios
- Users will be able to skip onboarding and complete their profile later
- Rippl will prompt users to connect via Open Banking or enter information manually during onboarding

Scenario Engine:

- Rippl will display default scenarios for users to choose from which will require temporary information from the user (mvp)
- Rippl will display a table comparing key financial figures from the user's current situation with the projected figures after the scenario has occurred. (mvp)
- Users will be able to save scenarios to "saved scenarios" (mvp)
- Users will be able to delete saved scenarios (post-mvp)
- Rippl will produce an AI summary alongside the comparison table to give an overview of the financial implications of the scenario and why they have occurred (mvp)
- Users will not be allowed to edit the saved scenario. If their financial information has changed they must update it in their profile (mvp)
- Users will be able to name their saved scenario (post-mvp)
- Users will be able to share a scenario to an external app (post-mvp)
- Rippl will produce a 12 month cash flow projection via bar chart in a saved scenario (mvp)
- Users will be able to run up to three scenarios at once (post-mvp)
- Rippl will create a 5 year projection view in a saved scenario (post-mvp)
- Rippl will allow users to input custom scenarios, not just default ones (post-mvp)
- Rippl will display a confidence or accuracy indicator on AI-generated projections (mvp)
- Rippl will allow users to duplicate a saved scenario as a starting point for a new one (post-mvp)
- Rippl will clearly indicate when a saved scenario's underlying profile data has changed and may be outdated (post-mvp)
- Users will be able to add notes to a saved scenario (post-mvp)
- Rippl will support currency selection for users outside the UK (post-mvp)

Notification & Life Alerts:

- Users can permit Rippl to send notifications (mvp)

- Users will be able to turn on “Life Alert” notifications which notify the user of a domestic or global impact on their finances (post-mvp)
- Users will be able to configure which categories of Life Alert they receive (e.g. interest rates, inflation, housing market) (post-mvp)
- Users will be able to set a threshold for alerts (e.g. only notify if impact exceeds £X) (post-mvp)
- Rippl will link Life Alert notifications to relevant saved scenarios where applicable (post-mvp)

Data & Privacy:

- Rippl will store user information in a secure PostgreSQL database (mvp)
- Users will be able to save their current financial information and other necessary information in their profile (mvp)
- Users will be able to upload a profile picture (post-mvp)
- Users will be able to update all the information in their profile except their email. (mvp)
- Users will be able to delete their accounts (mvp)
- Users will be able to export their data in a portable format (e.g. PDF or CSV) (post-mvp)
- Rippl will display a privacy policy and terms of service, which users must accept before creating an account (mvp)
- Rippl will allow users to request deletion of all their data in compliance with GDPR (mvp)
- Rippl will not share or sell user financial data to third parties (mvp)

Accessibility:

- Users will be able to switch to dark mode and switch back to light mode (post-mvp)
- Rippl will have different tabs that users can navigate between (mvp)
- Rippl will support dynamic text sizing (post-mvp)
- Rippl will meet WCAG 2.1 AA accessibility standards (mvp)
- All charts and visual outputs will include text-based alternatives for screen readers (mvp)

Open Banking: (post-mvp)

- Rippl will implement Open Banking via TrueLayer for accurate financial information
- Users will be able to connect and disconnect their bank account(s) via TrueLayer
- Rippl will display the date of the last Open Banking data sync
- Users will be able to manually trigger a data refresh
- Rippl will fall back gracefully if Open Banking connection fails, prompting manual entry
- Rippl will clearly indicate which profile fields are populated from Open Banking vs. manually entered

Family Mode: (post-mvp)

- Rippl will implement a family mode which models joint finances for scenarios
- Family mode will allow a primary user to invite one or more members via email or link
- Each family member will have their own profile and financial data
- The primary user will be able to remove members from a family group
- Family mode scenarios will clearly attribute income and expenses to individual members

- Users will be able to leave a family group

Reverse Mode: (post-mvp)

- Rippl will implement a “reverse mode” where Rippl will predict what finances a user needs for an intended outcome
- Users will be able to set a target outcome (e.g. afford a house, retire by 60) and Rippl will calculate required financial conditions
- Reverse mode results will be saveable in the same way as standard scenarios
- Rippl will indicate what changes to the user's current finances are required to meet the target

Error States & Edge Cases:

- Rippl will display meaningful error messages when a scenario cannot be calculated (mvp)
- Rippl will handle incomplete profile data gracefully, informing the user what is missing and why it's needed (mvp)
- Rippl will prevent scenario execution if required fields are missing and guide the user to complete them (mvp)

Non-Functional Requirements:

Performance:

- Rippl will load the home screen within 2 seconds on a standard 4G connection
- Scenario calculations will return results within 3 seconds for standard scenarios
- AI summaries will begin streaming or display a loading state within 1 second of request
- Charts and projections will render within 1.5 seconds of scenario completion
- Open Banking data sync will complete within 5 seconds or surface a timeout message

Scalability:

- Rippl's backend will be designed to handle a minimum of 10,000 concurrent users without degradation
- The database will be structured to support horizontal scaling as user volume grows
- The scenario engine will be stateless to allow parallel processing across multiple servers

Security: (mvp)

- All data in transit will be encrypted using TLS 1.2 or higher
- All data at rest will be encrypted using AES-256
- Rippl will comply with OWASP Top 10 security standards
- Authentication tokens will expire after 30 minutes of inactivity
- Open Banking credentials will never be stored by Rippl — only tokens issued by TrueLayer
- Rippl will undergo penetration testing before public launch
- All API endpoints will be protected against common attacks (SQL injection, XSS, CSRF)

- Rippl will implement rate limiting on authentication endpoints to prevent brute force attacks

Privacy & Compliance: (mvp)

- Rippl will comply with GDPR for all UK and EU users
- Rippl will comply with FCA regulations applicable to financial data display and Open Banking
- Rippl will maintain an audit log of all access to user financial data
- User consent will be obtained and recorded before any data processing begins
- Rippl will comply with Apple App Store and Google Play Store data privacy policies

Reliability & Availability:

- Rippl will target 99.9% uptime (equates to ~8.7 hours downtime per year)
- Rippl will implement automated health checks and restart failed services
- Rippl will have a defined disaster recovery plan with a Recovery Time Objective (RTO) of under 4 hours
- Rippl will degrade gracefully — if AI or Open Banking services are unavailable, core features will still function

Maintainability:

- Rippl's codebase will follow a consistent style guide enforced by a linter
- All core business logic will have minimum 80% unit test coverage
- The codebase will be modular to allow independent updates to the scenario engine, AI layer, and Open Banking integration
- All API endpoints will be versioned to allow backward-compatible updates

Usability:

- Rippl will be usable by a financially literate adult with no prior training
- All core user journeys will be completable in no more than 5 taps from the home screen
- Rippl will conform to platform UI conventions (iOS Human Interface Guidelines / Android Material Design)
- Error messages will be written in plain English with a clear next action
- Rippl will be available in different languages

Compatibility:

- Rippl will support iOS 16 and above
- Rippl will support Android 10 and above
- Rippl will be optimised for screen sizes from 4.7 inches to 6.7 inches
- Rippl will support both portrait and landscape orientations

AI Reliability:

- AI-generated summaries will include a disclaimer that outputs are informational and not financial advice
- Rippl will log AI inputs and outputs for quality monitoring and auditing purposes

- AI outputs will be reviewed against a set of guardrails to prevent financially misleading statements

Notifications:

- Push notifications will be delivered within 60 seconds of a triggering event
- Life Alert notifications will include a source reference for the event that triggered them

Assumptions & Limitations:

Assumptions

Users

- Users are financially literate enough to understand basic terms (income, expenses, cash flow, net worth)
- Users will have access to a smartphone running iOS 16+ or Android 10+
- Users are based primarily in the UK at launch
- Users have at least one active UK bank account for Open Banking integration
- Users are willing to share financial data in exchange for personalised projections
- Users who connect via Open Banking will have accounts supported by TrueLayer's network

Market

- There is sufficient demand for a personal financial scenario planning tool at the price point chosen
- The waitlist of 200 will be representative of the broader target market
- Competing apps will not launch a directly equivalent feature set during the MVP development window

Technical

- TrueLayer will maintain its current API availability and pricing structure
- The AI model used for summaries will remain accessible, affordable, and consistent in output quality at scale
- Apple and Google will approve the app without requiring significant changes to core features
- Push notification delivery rates will remain consistent across iOS and Android platforms
- Open Banking data provided via TrueLayer will be sufficiently accurate and up to date for projection purposes

Regulatory

- Rippl's scenario output will be classified as a financial tool, not financial advice, avoiding FCA authorisation requirements at MVP
- GDPR compliance will be achievable within the development timeline

- TrueLayer holds the necessary FCA permissions, covering Rippl's Open Banking data access

Business

- Pro subscription revenue will be the primary monetisation model at launch
- The B2B/IFA white-label opportunity in Year 2 will not require a fundamental rebuild of the core product

Constraints

Time

- MVP must be buildable within a defined solo development timeline alongside other commitments
- Waitlist validation must be completed before any significant build investment is made

Budget

- Development will be self-funded initially, limiting the use of paid third-party services at scale
- AI API costs must remain commercially viable at the projected subscriber volumes — cost per AI summary call must be modelled early
- TrueLayer's pricing structure will impose a per-connection cost that affects unit economics

Technical

- Rippl is a mobile-only product at launch — no web version will be built in Year 1
- Family mode complexity means it is a Post-MVP feature and will require significant additional architecture
- The scenario engine is limited to modelling scenarios with quantifiable financial inputs — lifestyle or behavioural factors cannot be modelled
- Reverse mode requires significantly more computational logic than standard scenarios and is Post-MVP

Legal & Regulatory

- Rippl must not present AI outputs as regulated financial advice under FCA guidelines
- All user data must be stored within UK/EU jurisdiction to comply with GDPR
- Rippl cannot store raw Open Banking credentials under any circumstances

Platform

- Apple and Google review processes may delay launch timelines unpredictably

- iOS and Android platform differences may require separate handling of certain features (e.g. biometrics, notifications)

Open Questions:

- Will we launch Rippl globally or stay UK-based?
- Will we keep Rippl public or pitch to businesses and firms?
- Will we eventually sell Rippl or keep it forever?
- Will we make a website for Rippl?
- How will Rippl handle users whose bank is not supported by TrueLayer?
- What is the fallback if the AI provider changes pricing or deprecates the model being used?
- How will family mode handle a scenario where one member deletes their account?